

MUHAMMAD USMAN AL HAQ

Software Engineer — Backend & Cloud (.NET / Azure) | DevOps & Infrastructure | Computer Science Senior @ FAST-NUCES

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PROFESSIONAL SUMMARY

Computer Science senior at FAST-NUCES Islamabad focused on backend and cloud engineering with .NET and Azure. Built and deployed ServiceDesk, a multi-tenant IT service management platform on Azure Container Apps, using PostgreSQL Row-Level Security for tenant isolation, Dockerized services, Hangfire background jobs, and a GitHub Actions CI/CD pipeline. ICPC Asia Regionalist with a strong foundation in systems programming, having benchmarked CPU+GPU co-execution pipelines in CUDA/AVX2/OpenMP. Currently a remote full-stack intern at Sigma Evolution. Looking to grow into DevOps/MLOps through a part-time or internship role alongside university.

TECHNICAL SKILLS

Cloud & DevOps: Microsoft Azure (Container Apps, Azure Fundamentals certified), Docker, CI/CD (GitHub Actions), Hangfire background jobs, OpenTelemetry / Application Insights, Vercel

Languages: C#, C++, Python, TypeScript, JavaScript, Kotlin, Java, PHP, SQL, HTML/CSS

Backend & Frameworks: ASP.NET Core 8, Entity Framework (EF) Core, Node.js, Next.js, React, JavaFX, Android SDK

Databases: PostgreSQL (Row-Level Security / multi-tenant), MySQL, SQL Server, MongoDB, Firebase Firestore, SQLite, Supabase

Tools & Platforms: Git/GitHub, Postman, Figma, Linux, VS Code, Android Studio, XAMPP

Concepts: Multi-Tenant Architecture, RBAC, RESTful APIs, MVC, Parallel & Distributed Computing, OOD, Agile

PROFESSIONAL EXPERIENCE

Full-Stack Intern — Sigma Evolution (Remote)

June 2026 – Present | Pakistan

- Set up and manage the cloud database for the team's hospital management web platform (React, Node.js/Express, FastAPI, MySQL on Aiven), built with a student-run remote team.
- Built a GitHub Actions workflow that keeps the free-tier Aiven database alive — detecting and restarting it when idle and running a scheduled keep-alive query to avoid downtime.

Web Development Intern — Kuwait Kazmah Center

June 2025 – Aug 2025 | Hawally, Kuwait

- Redesigned and restructured the company website (kkc.com.kw) end-to-end using WordPress, improving user experience and visual consistency.
- Created bilingual (English & Arabic) customer feedback forms to streamline review collection.
- Migrated company files to Microsoft Access and built queries to speed up data retrieval; launched and managed the company's Instagram page.

FEATURED PROJECT — CLOUD & DEVOPS

ServiceDesk — Multi-Tenant IT Service Management Platform | ASP.NET Core 8, React (Vite), PostgreSQL, EF Core, Docker, Azure Container Apps — [GitHub](#)

- Architected a full-stack ticketing/ITSM platform with role-based access control and multi-department tenant isolation enforced at the database layer via PostgreSQL Row-Level Security.
- Deployed the ASP.NET Core 8 Web API to Azure Container Apps with a Dockerized build, and shipped a GitHub Actions CI/CD pipeline for automated builds and releases.
- Implemented automated SLA tracking and background processing with Hangfire, plus OpenTelemetry/Application Insights instrumentation for tracing and metrics.
- Wrote integration tests with Testcontainers to spin up isolated PostgreSQL instances per test run, verifying RLS policies behave exactly as in production; built a React + TailwindCSS SPA frontend and a complete audit trail.

OTHER PROJECTS

FastPay — Enterprise Digital Wallet & P2P Ledger Ecosystem | Java 17, JavaFX, Next.js 14, Supabase, PostgreSQL, TypeScript

- Engineered a full-stack fintech ecosystem — JavaFX desktop client paired with a Next.js 14 web portal — backed by an ACID double-entry ledger, JWT auth, and Postgres Row-Level Security.
- Delivered virtual budget partitions, P2P/Raast fund transfers, QR payment slips, and a real-time searchable grid of 26+ utility billers.

Hybrid CPU+GPU Co-Execution for 2D Image Convolution | C++, CUDA, OpenMP, AVX2 SIMD, Ubuntu

- Implemented four variants of 15×15 separable Gaussian convolution — sequential, CUDA shared-memory, AVX2+OpenMP, and a hybrid co-execution pipeline.
- Profiled performance from 512×512 to 4K resolution, identified a 40% PCIe transfer bottleneck, and proved the hybrid pipeline outperforms an isolated GPU pipeline.

CineBook Pro — Cinema Ticket Booking Platform | Next.js 14, MongoDB, NextAuth.js, TMDB API, Tailwind CSS

- Built a full-stack cinema booking platform with interactive seat maps, a role-based admin panel, and automated movie import via the TMDB API; secured with bcrypt hashing, token-based password resets, and JWT sessions.

COMPETITIONS & ACHIEVEMENTS

- ICPC Asia Topi Regionalist 2025 — represented FAST Islamabad at GIK Institute, clearing the qualifying round among top teams nationwide.
- Led a tree-planting initiative with the Red Crescent Society Islamabad (70+ trees).

EDUCATION

Bachelor of Science, Computer Science — FAST-NUCES

Aug 2023 – Present | Islamabad, Pakistan

HSSC I & II, Pre-Engineering — International School of Pakistan

2020 – 2022 | Kuwait — Score: 957/1100 (FBISE)

CERTIFICATIONS & LANGUAGES

Certifications: Essentials with Azure Fundamentals; Develop an ASP.NET Core Web App that Consumes an API; Strategic Analysis and Solution Design; Computer Fundamentals

Languages: English (Full Professional), Urdu (Native), Hindi (Professional), Arabic (Elementary)